VORTEX SHEDDING IN BLUFF BODIES

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Introduction

The interaction of fluid flow with solid structures is a fundamental concept in fluid mechanics, with profound implications in fields ranging from civil and aerospace engineering to biomechanics. Structures can be broadly classified as either streamlined or bluff, based on their geometric profile and the resulting flow characteristics. Unlike streamlined bodies, which are designed to minimize fluid resistance, **bluff bodies** (such as cylinders, spheres, or flat plates) are characterized by shapes that cause significant **flow separation** from their surfaces.

A critical and highly studied phenomenon associated with fluid flow past a bluff body is **vortex shedding**. This occurs when the flow separates from the body, forming alternating vortices (swirling regions of fluid) that are shed periodically from opposite sides of the structure. This periodic shedding creates a distinct wake pattern known as a **Kármán vortex street**.

The primary objective of this experiment is to investigate and quantify the frequency of vortex shedding in the wake of bluff bodies. By measuring this frequency for different body geometries, this investigation aims to validate the established dimensionless relationship between flow velocity, body size, and shedding frequency, which is characterized by the **Strouhal number (St)**.

Theoretical Background: Bluff Bodies and Flow Separation

Defining a Bluff Body

A **bluff body** is an object whose geometry is not streamlined in a way that aligns with the oncoming fluid flow. The defining characteristic of a bluff body is that the dominant component of its total drag is **pressure drag** (also known as form drag), rather than skin friction drag. This high pressure drag is a direct and defining consequence of large-scale flow separation.

Flow Separation and Wake Formation

The behavior of fluid flowing past a bluff body is heavily dependent on the **Reynolds number (Re)**, a dimensionless quantity that compares the fluid's inertial forces to its viscous forces.

- At very low Reynolds numbers (creeping flow, Re < 5), the fluid's viscosity is dominant. The flow remains attached to the entire surface, moving smoothly around the object in what is known as Stokes flow. No separation occurs.
- As the Reynolds number increases (Re= 10-40), the flow begins to separate from the rear surface of the body, forming a pair of stable, stationary vortices in the near-wake.
- Upon reaching a critical Reynolds number (Re = 47 for a circular cylinder), this stable wake becomes unstable. The vortices begin to detach from the body, one at a time, from alternating sides. This is the onset of the Kármán vortex street.

This **flow separation** occurs because the fluid particles, having travelled from the high-pressure front (stagnation point) to the low-pressure sides, do not have enough momentum to overcome the adverse pressure gradient on the rear half of the object. The flow "peels" away from the surface, creating a broad, turbulent, low-pressure region behind the body known as the **wake**. This significant pressure difference between the high-pressure front and the low-pressure wake is the primary source of the large pressure drag.

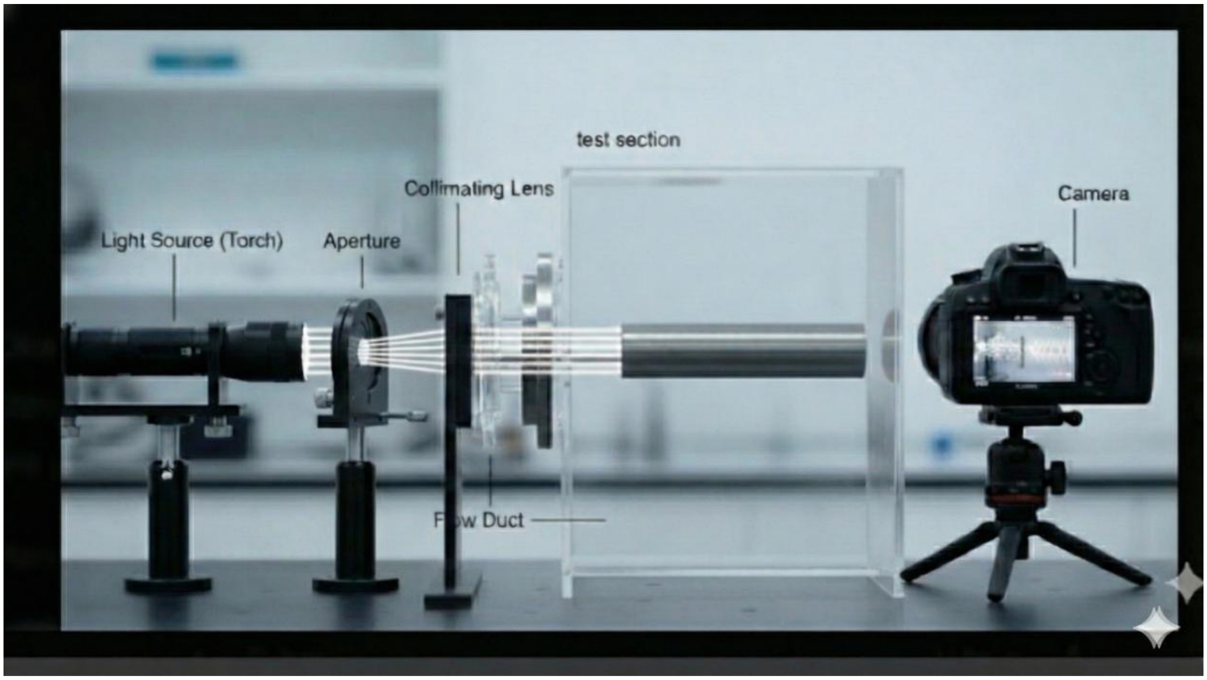
Common examples of bluff bodies relevant to engineering include:

- Circular cylinders (e.g., chimneys, pipelines, bridge cables)
- Spheres (e.g., sports balls)
- Flat plates oriented perpendicular to the flow
- Square or rectangular beams (e.g., building structures)

The specific geometry of the bluff body (e.g., the sharp corners of a square beam versus the smooth curve of a cylinder) dictates the exact point of flow separation, which in turn governs the characteristics of the wake and the frequency of the vortices shed.

Experimental Setup

To observe and record the vortex shedding phenomenon, a non-intrusive flow visualization technique known as **shadowgraphy** was employed. This method is highly effective at visualizing flow features that involve gradients in fluid density, such as the vortices in the wake of a bluff body.



The apparatus, shown schematically in Figure 1, was arranged in a linear configuration.

Figure 1: A schematic of the shadowgraph optical setup used to visualize and capture the vortex street shed from a bluff body within a flow duct.

The key components of the setup are as follows:

1. **Illumination System:** The light path originated with a high-intensity light source (a torch). The light was immediately passed through an aperture, which served two critical functions:
    a. It created a more coherent, point-like source of light, which is essential for generating sharp shadow images.
    b. It allowed for the control of the overall brightness and intensity of the illumination.
2. **Collimating Lens:** The divergent light rays exiting the aperture were then directed into a collimating lens. This lens was precisely positioned to refract the light into a parallel beam, ensuring that all light rays traveled parallel to one another through the test section.
3. **Test Section (Vertical Duct):** The parallel light beam illuminated a transparent **vertical duct**. The bluff body, a **circular disc**, was mounted horizontally inside this duct, with its flat face oriented **normal (perpendicular) to the direction of the upward airflow**. A flow source (not shown) provided a steady upward flow of fluid through the duct.

4. **Visualization and Recording:** As the fluid flowed upwards past the circular disc, it generated a periodic Kármán vortex street in the wake above the disc. These vortices are regions of slightly different fluid density and temperature compared to the surrounding freestream flow. Based on the principles of shadowgraphy, these density gradients refract the parallel light rays that pass through them.
5. **Camera:** A camera was positioned horizontally, looking through the side of the vertical duct. It captured the pattern of light and dark areas (shadows) created by the refracted light, providing a direct visualization of the vortex shedding forming above the disc. This video footage was recorded for subsequent frequency analysis.

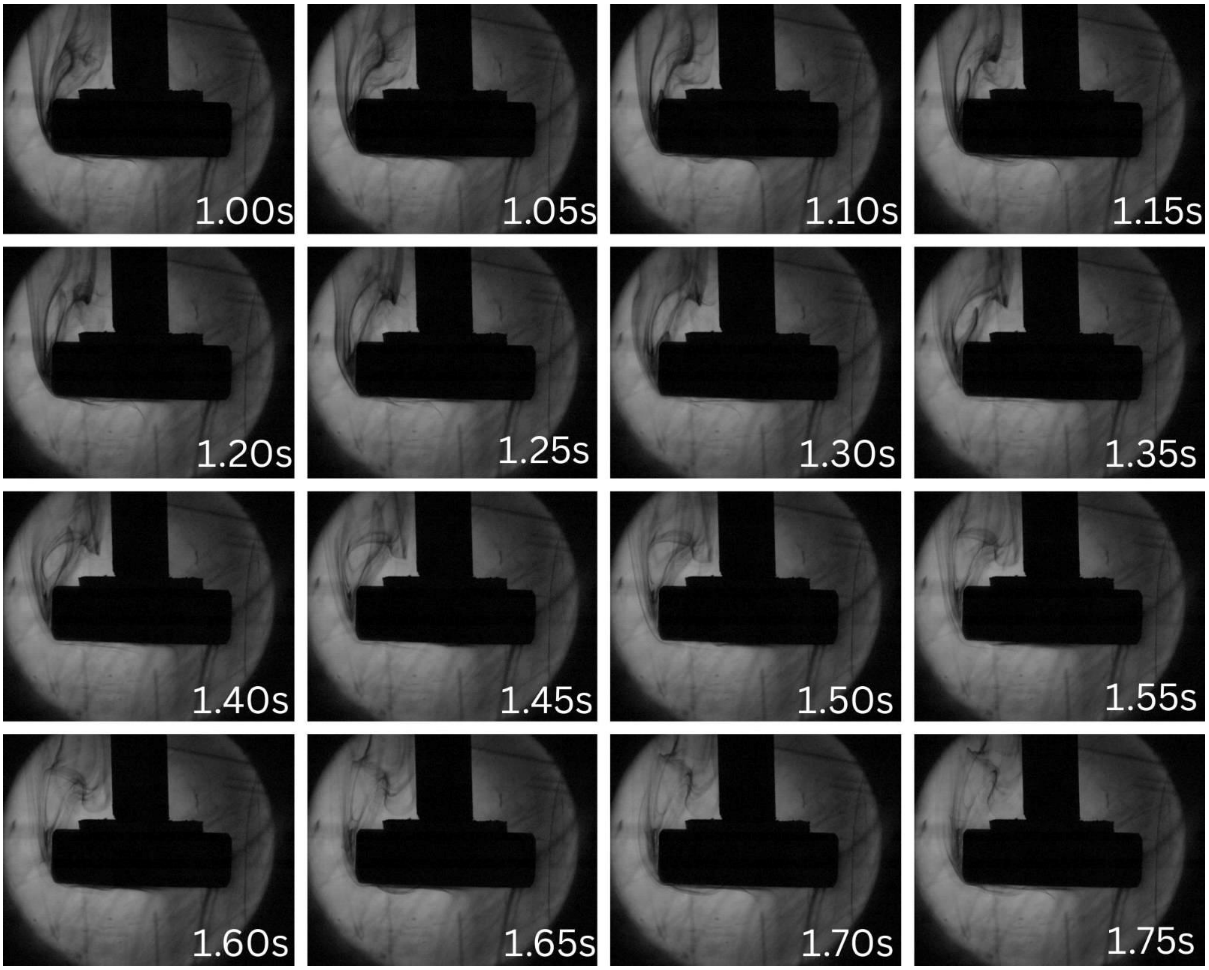
Experimental Observations and Results

The experiment conducted successfully demonstrated the classic fluid dynamics phenomenon of vortex shedding using shadowgraphy techniques. The main objective was to observe the behaviour of airflow around a bluff body, identify the characteristic wake pattern known as a von Karman vortex street.

Flow Visualization and Wake Formation

The smoke stream was initiated, and a smooth, columnar flow was established such that it went past the bluff body. As this flow reached the body, it was forced to separate at its sharp upstream edges. The separation took place in the form of a wide, disturbed region, which was directly in the wake of the body. The flow in this wake was turbulent initially, but it quickly turned into a stable and repetitive pattern.

The most significant result of the experiment was the clear formation of a von Karman vortex street which lasted for an extended period. It indicated a consistent, periodic shedding of swirling vortices from both the upper and lower surfaces of the body.



Qualitative Analysis of Vortex Shedding

The shedding process took place at regular intervals and was also a continuous one. The vortices retained their coherent, rotating structure until gradually dissipating through the action of viscosity and mixing. The size of each vortex was about the same and was approximately equal to the characteristic dimension of the bluff body. Since the observed flow pattern was stable it clearly indicated that a steady state was achieved.

The study was mainly intended to be qualitative however the shedding was so regular that it became possible to estimate a fundamental frequency. A dominant shedding rhythm of 1.25 Hz was measured by observing the release of the vortices over time. This frequency caused the smoke trail to have a continuous, wavy, and undulating shape, which gave a clear visual rhythm to the otherwise complicated fluid motion.

Thus, the experiment gave a clear visualization of vortex formation and shedding. The change from smooth flow to an oscillating wake showed the unsteady behaviour of bluff-body separation. The 1.25 Hz rhythm observed is a result of the interaction between the flow's inertia and the obstruction presented by the body. This is a clear measurable indicator of this fundamental process.

Results

- The series of 16 time-lapse images clearly captures the vortex shedding phenomenon.
- Typical von Karman vortex street with periodic formation and detachment of alternating vortices in the wake of the bluff body can be clearly seen from these images.

- The dominant frequency of the shedding cycle was measured and was found to be 1.25 Hz. This was done by tracking the progression of these structures throughout the image sequence.

- The frequency experimentally found is 1.25Hz and it is related to the flow velocity and the body geometry.

- The oscillation that was seen is due to the instability of the flow when it separates from the body's surface. The regularity of the flow in the images confirms the Strouhal number's ability to predict this common fluid dynamic event.

Conclusion

The experiment successfully validated the theoretical predictions of vortex shedding behavior behind bluff bodies, demonstrating strong agreement between observed phenomena and established fluid mechanics principles. The shadowgraphy technique effectively captured the formation of a distinct von Kármán vortex street in the wake of the circular disc, with vortices shedding periodically from alternating sides at a measured frequency of 1.25 Hz.

The observed vortex shedding pattern closely matched theoretical expectations based on Reynolds number considerations. The regular, alternating formation of coherent vortices confirmed that the flow regime exceeded the critical Reynolds number (Re = 47 for circular cylinders), where wake instability transforms stable vortices into periodic shedding. The measured shedding frequency of 1.25 Hz. This dimensional consistency between experimental frequency, body dimensions, and flow velocity validates the fundamental fluid-structure interaction principles governing bluff body aerodynamics.

The shadowgraphy setup proved highly effective for non-intrusive flow visualization, successfully revealing density gradients associated with vortex structures. The experimental observations demonstrated key characteristics of bluff body flow separation: the transition from initially turbulent wake conditions to a stable, repetitive shedding pattern, the formation of coherent vortices with dimensions proportional to the body size, and the sustained periodicity indicating steady-state flow conditions. The pressure drag resulting from large-scale flow separation and the subsequent low-pressure wake region aligned with theoretical predictions for bluff bodies, where form drag dominates over skin friction. These results provide quantitative validation of fundamental concepts in fluid mechanics and demonstrate the practical applicability of dimensionless analysis in predicting real-world flow phenomena with engineering significance for structures such as chimneys, bridge cables, and building components exposed to fluid flows